webSPDZ · ATLAS demo — easy start guide

No coding, no typing commands into a browser, no technical knowledge needed. Follow the numbered steps in order. This guide assumes you are starting from nothing — just a computer with internet access.

What is this? A small demo of "private computation" — up to 4 people each type in 4 private numbers, and the computer works out who should be paired with whom, *without anyone's numbers ever being shown to anyone else* — not even to the computer doing the matching. This guide runs the whole demo on **one computer**, in 4 separate browser tabs (simulating 4 people).

1 Install Docker Desktop (one-time setup)

Docker is a free program that lets this demo run the same way on any computer. You only need to install it once.

1. Open your web browser and search for **"Docker Desktop download"**, or go to `docker.com` and look for "Docker Desktop."
2. Download the version for your computer:
   - **Mac with Apple Silicon** (M1/M2/M3/M4 chip, most Macs sold since 2021) — choose "Mac with Apple chip."
   - **Mac with Intel chip** (older Macs) — choose "Mac with Intel chip." Not sure which you have? Click the Apple menu (top-left corner) → **About This Mac** — it tells you.
   - **Windows** or **Linux** — choose the matching installer.
3. Open the downloaded file and follow the installer — keep clicking the default/suggested option until it finishes.
4. Once installed, open the **Docker Desktop** application (search for it like any other app). Wait until it says it's running — there's usually a small whale icon in your menu bar / system tray once it's ready.

You will not need to open Docker Desktop itself again after this — just leave it running in the background. It quietly does its job once installed.

2 Get the project folder

You should have received a folder (or a `.zip` file) called **webSPDZ**. If it's a `.zip`, double-click it to unzip it — on Mac this happens automatically; on Windows, right-click → **Extract All**.

Put the unzipped **webSPDZ** folder somewhere easy to find, like your Desktop.

3 Open a Terminal in that folder

The Terminal is just a window where you type one line of text. You'll only type one thing, once.

On a Mac:

1. Open the **webSPDZ** folder in Finder.
2. Right-click (or Control-click) anywhere empty inside the folder window.
3. Choose **New Terminal at Folder** from the menu. (If you don't see this option: open the **Terminal** app from Applications → Utilities, then type `cd` — with a space after it — drag the webSPDZ folder into the Terminal window, and press Enter.)

On Windows: open the **webSPDZ** folder, click the address bar at the top, type `cmd`, and press Enter.

4 Start the demo

In the Terminal window that opened, type exactly this and press Enter:

```
docker compose up --build
```

You'll see a lot of text scroll by — that's normal. The first time takes under a minute. When it settles down and stops scrolling, it's ready. **Leave this Terminal window open** for as long as you're using the demo — closing it stops everything.

5 Tell your browser to trust this page (one-time per session)

This demo runs on your own computer only, but your browser doesn't know that automatically, so it shows a warning the first time. This is expected and safe to click through — you are only connecting to your own computer, nothing goes out to the internet.

Open **Google Chrome** and visit these two addresses, one at a time. On each one:

1. You'll see a page saying something like *"Your connection is not private."*
2. Click **Advanced** (small text, usually bottom-left).
3. Click **Proceed to localhost (unsafe)**.

Visit both of these once, clicking through the warning each time:

```
https://localhost:8000
https://localhost:8080
```

6 Open the demo — 4 browser tabs, same address

Open **4 new tabs** in Chrome (one for each pretend "participant"). In **every** tab, go to the exact same address:

```
https://localhost:8000/atlas-lobby.html
```

That's it — the same single link, four times. No long addresses to copy, nothing to type into the address bar beyond this.

7 What you'll see — the waiting room

Each tab shows a small grid of 4 "seats," like this:

Seats

P0 (you)	P1	P2	P3
Joined	Joined	Empty	Empty

As each tab loads, it claims the next empty seat automatically — you don't pick one. The grid updates live in every tab as more tabs join.

8 Type in 4 numbers, in every tab

Each tab asks for 4 numbers (think of them as 4 "skill levels," one per box):

Your private input



Type any 4 whole numbers you like into each of the 4 boxes, in each tab — they don't have to match between tabs, and in fact they shouldn't. A simple way to think about it: the number 10 is "neutral." Numbers above 10 mean "I have extra of this to share," numbers below 10 mean "I need more of this." Then click **I'm ready** in that tab.

Do this in all 4 tabs. Nothing starts until every tab has clicked ready — feel free to take your time on each one.

Your numbers never leave that browser tab. Clicking "ready" only tells the other tabs "I'm done typing," not what you typed. The matching happens through encrypted computation between the tabs directly — that's the whole point of this demo.

9 Reading the result

Once all 4 tabs have clicked ready, each tab automatically shows its own private result a few seconds later — nobody needs to click anything else:

What you see	What it means
P followed by a number (e.g. "P2")	This tab's best match was whichever tab holds that seat number.
No match	Nothing useful was found to pair this participant with, based on the numbers typed in.

Each tab only ever shows *its own* result — that's the "private" part of private computation.

10 Stopping the demo

When you're done, go back to the Terminal window from Step 4 and press **Ctrl** + **C** (hold Ctrl, tap C). That shuts everything down. You can close the Terminal window after that.

To run it again later: repeat Step 4 (`docker compose up --build` in that same folder) and Step 6 — Steps 1–3 and 5 don't need repeating.

11 If something doesn't look right

What happened	What to do
Terminal says something about a "port" already being used	Something else on your computer is already using the demo's address. Restart your computer and try again — or ask whoever set this up for you for help.
A tab says **"Room is full"**	More than 4 tabs tried to join at once. Close the extra tab(s) — only 4 are needed.
A tab is stuck on "Joined" and nothing happens after clicking ready	Check that all 4 tabs clicked ready — the grid in each tab shows which seats say "Ready" vs "Joined." It only starts once all 4 say "Ready."
You accidentally closed one of the 4 tabs	Open a new tab to the same address (Step 6) and start over from Step 8 in all 4 tabs — the seat that closed becomes empty again for the new tab to take.
Nothing loads at all / blank page	Make sure the Terminal window from Step 4 is still open and hasn't shown any red error text. If it has, close it, reopen a Terminal (Step 3), and run Step 4 again.